



Конспект «Операции с данными в SQL»

1. Простые запросы: ALIAS

```
SELECT f.title, c.name, f.rental_rate/f.rental_duration AS  
cost_per_day  
FROM film f  
JOIN film_category fc ON fc.film_id = f.film_id  
JOIN category c ON c.category_id = fc.category_id;
```

2. Простые запросы: ORDER BY

```
SELECT title, rental_rate/rental_duration AS cost_per_day  
FROM film  
ORDER BY cost_per_day DESC, title;
```

3. Простые запросы: LIMIT и OFFSET

```
SELECT title, rental_rate/rental_duration AS cost_per_day  
FROM film  
ORDER BY cost_per_day DESC, title  
LIMIT 10  
OFFSET 57;
```

4. Простые запросы: DISTINCT

```
SELECT DISTINCT first_name  
FROM customer;
```

5. Простые запросы: WHERE

```
SELECT *  
FROM payment  
WHERE amount > 7 AND staff_id = 2 OR amount < 5 AND staff_id = 1;
```

6. Простые запросы: CAST

```
SELECT payment_id, CAST(payment_date AS DATE)  
FROM payment;
```

7. Округление

```
SELECT ROUND(100.576); -- 101  
SELECT ROUND(100.576, 2); -- 100.58  
SELECT TRUNCATE(100.576, 2); -- 100.57  
SELECT FLOOR(100.576); -- 100  
SELECT CEIL(100.576); -- 101  
SELECT ABS(-100.576); -- 100.576
```

8. Арифметические операторы

```
SELECT POWER(2, 3); -- 8  
SELECT SQRT(64); -- 8  
SELECT 64 DIV 6; -- 10  
SELECT 64%6; -- 4  
SELECT GREATEST(17, 5, 18, 21, 16); -- 21  
SELECT LEAST(17, 5, 18, 21, 16); -- 5  
SELECT RAND(); -- 0.005757967015502944
```

9. Работа со строками

```
SELECT CONCAT(last_name, ' ', first_name, ' ', email) FROM  
customer;  
SELECT CONCAT_WS(' ', last_name, first_name, email) FROM  
customer;
```

10. Работа с датами и временем

```
SELECT DATE_ADD(NOW(), INTERVAL 3 DAY);
```

```
SELECT DATE_SUB(CURDATE(), INTERVAL 3 DAY);
```

```
SELECT YEAR(NOW()), MONTH(NOW()), WEEK(NOW()), DAY(NOW());
```

11. BETWEEN

```
SELECT * FROM payment WHERE amount BETWEEN 5 AND 7;
```